

Exact Product Capture under a Finite Single-Port Parent Store for Event-Driven Spiking Optical Flow

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Abstract—Low firing activity does not automatically accelerate event-driven optical flow when parent-state traffic is bound by finite storage and a single 1RW port. This brief reports one exact product-capture island, C1, for a frozen binary-event optical-flow checkpoint. A row may reuse only a still-resident exact-subset parent, issues the XOR residual, and reconstructs the signed product row by addition. Dead writes that no later row can observe are elided. Completion of a row and its final psum write are atomic under a deadline-aware 1RW arbiter. On 51.84 million source rows from ten *zurich_city_09_a* samples, the same-ledger cycle model reduces component time by 40.99% (1.6945 $\times$) versus strongest-zero skipping; same-coordinate bit skipping on that ledger is only 1.003 $\times$. The mapped 28-nm nine-SRAM island occupies 166,514 μm^2 , meets 3-ns setup/hold, dissipates 29.08 mW / 22.07 nJ in a 64-row/253-cycle directed window, and passes 16,549 mapped-to-mapped Formality compare points. A real-mask VCS tile calibrates event counts; the 1.6945 $\times$ ratio remains a cycle-model result. All physical numbers are prelayout and pre-macro-extracted. We make no whole-network speedup claim.

Index Terms—Digital accelerator, event-based vision, optical flow, product sparsity, single-port SRAM, spiking neural networks.

EVENT cameras make inactivity visible, but sparse arithmetic is only one term in execution time. In a binary-event optical-flow deployment, nonzero sources still trigger parent-state reads, residual issues, and psum writes. A design that reports only operation sparsity can therefore be slower than an equal-resource zero-skipping baseline once the parent working set is finite and the store is single-ported.

Prosperity [1] and Phi [2] show that useful SNN speedups come from executable reuse or patterns, not firing rate alone. Digital neuromorphic processors such as ODIN [3] and event-stream accelerators such as SpinalFlow and SNE [4], [5] establish spike-driven datapaths. TCAS-II SRAM compute-in-memory macros exploit sparsity inside the array [6], [7]. Our object is different: repeated *products* of a convolution-like layer, subject to a 240-KiB-class parent lifetime and a physical 1RW contract.

This brief makes one architectural contribution. C1 captures exact repeated products while a legal parent remains resident, reconstructs the child from the residual, and never exposes a partial row. The frozen Motion checkpoint is an evaluation workload, not a parallel algorithmic claim. A companion typed weight-broadcast fabric for fully connected layers exists in the

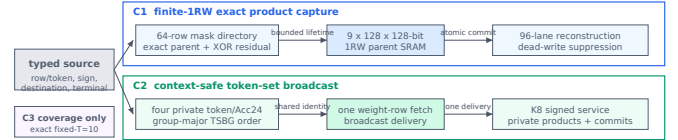


Fig. 1. Typed-source context of the frozen optical-flow network. This brief reports only C1: exact product capture in a finite 1RW parent store. The other branches are workload context and are not claimed.

same project; its hold and power are not closed, so it is not claimed here.

I. WORKLOAD AND EXACTNESS CONTRACT

The frozen Motion C12 ep34 checkpoint obtains 1.199514 average endpoint error (AEE) on an 825-frame local DSEC validation set and has a 5.6709% global firing rate. Of 105 configured ATLIF wrappers, 12 *sn2_q* wrappers are runtime-bypassed, so capture invokes 93 (48 $T=2$, 45 $T=10$). A graph audit finds that 12 invoked *attn_sn* returns have no consumer under fixed normal inference, leaving 81 graph-live services. All 93 captured ATLIF outputs are binary. Signed source codes are a downstream execution-protocol property (polarity and correction traffic), not evidence that the captured tensors are analog. The frozen software path uses Motion-XOR scoring with $\alpha=0.125$ and K as its value carrier; hardware quantization is disabled on that path.

Each result is assigned one evidence class. A cycle-model result replays the same trace and charged resources in software. A VCS result proves directed RTL function and protocol properties. DC/PT/Formality results anchor area, timing, and equivalence. Only a result that closes all three classes on one workload may be called an RTL speedup; this brief reports the three classes side by side.

Bottleneck convolution exposes repeated products only while parent and destination state remain resident. That invariant defines C1. Attention and decoder traces remain part of the frozen identity but are not promoted to performance contributions.

II. C1 MICROARCHITECTURE

A. Directory and Legal Parents

C1 first builds a ping-pong directory for each 64-row, 16-source mask task. A row may select only an earlier exact-subset row as its parent. Stable population-count ordering

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makes that choice unique: among legal parents the engine prefers the earliest still-live candidate so that later rows keep a deterministic residual. A live-parent bitmap records which completed rows remain observable. When the bitmap shows that no subsequent row can read a stored psum, that location is a dead write and is elided exactly rather than being scheduled as a drain.

The 240-KiB-class working set is the capacity that must be respected, not a DRAM-backed scratchpad. Nine 128×128 1RW SRAM macros hold parent masks and psums. The engine never assumes the ideal simultaneous 1RW access that produced a higher but unphysicalized reuse ceiling. This is the object difference from an unbounded product-sparsity opportunity: reuse must survive capacity, port, parent lifetime, and completion constraints.

B. Residual Issue and Exact Reconstruct

A selected parent mask $\mathcal{M}_p \subset \mathcal{M}_r$ yields a disjoint residual. C1 issues only that residual and reconstructs the 96-lane signed product row by adding the stored parent:

$$\mathbf{P}_r = \mathbf{P}_p + \sum_{s \in \mathcal{M}_r \setminus \mathcal{M}_p} \mathbf{p}_s. \quad (1)$$

Exactness follows from the disjoint union, not from value prediction. If no resident legal parent exists, C1 selects the empty parent and issues the original source row. Sign, destination, and accumulator identity are never coalesced across rows. Thus sparsity changes delivered work without changing the operator result.

C. Single-Port Arbiter and Atomic Commit

The execution side combines one earliest-parent lookahead, a two-entry reserved response queue, and forwarding under a deadline-aware 1RW arbiter. A parent read, a residual product, and a psum write compete for the same macro port. The arbiter grants the access that would otherwise miss its row-completion deadline; the reserved queue absorbs one already-accepted response so that a later grant cannot orphan it. Forwarding supplies a just-written parent to the next residual without a second macro read when the bitmap still marks that parent live.

Final psum write and row completion are a single atomic event. Backpressure may change service order, but it cannot expose a partially reconstructed row: the typed-source terminal bit and the completion pulse retire together. A directed VCS tile on the first 64 real support masks of the cycle ledger is used later only to calibrate these event counts, not to replace the same-ledger cycle model.

Speculative parent hits are disallowed. A candidate is legal only if its mask is an exact subset and the live-parent bitmap is still set at grant time, not at directory build time. That second check is what makes (1) exact under eviction: a parent that died between directory construction and issue is treated as missing, and C1 falls back to the empty parent. The fallback is slower than a stale hit and is the price of never reconstructing from a psum that the store no longer owns.

TABLE I
C1 SAME-LEDGER CAPTURE LADDER (51.84 M SOURCE ROWS).

Execution point	Cycles (M)	vs. zero	Boundary
Strong zero	648.741	1.000×	baseline
Same-coordinate bit	646.619	1.003×	baseline
C1 finite 1RW	382.849	1.6945×	cycle model
Concurrent-access	341.058	1.902×	ceiling only

III. EVALUATION

All physical results use a commercial 28-nm flow at 3.0 ns. Reported points are prelayout, use ideal clocks and ZeroWireload, and have no extracted parasitics. C1 area includes the nine SRAM macros. Logic-only comparisons from other islands in the same project are not mixed into Table III.

Table I exposes the baseline ladder instead of attributing all inactivity to product capture. The ledger is 51.84 million source rows from ten zurich_city_09_a ep34 samples. Per-sample ratios span 1.661–1.723×, with a 1.6946× geometric mean. Strongest-zero skipping still walks parent-state traffic for every nonzero source; same-coordinate bit skipping on this ledger removes almost none of that traffic (1.003×). Exact parent reuse is what creates the material reduction. The remaining 12.25% gap from C1 to the concurrent-access ceiling is the tax of the physical single-port contract. We do not promote the ceiling as an attainable RTL result.

The cycle model charges the same resources the RTL would charge: one 1RW grant per accepted macro access, residual issue, forward, deadline hold, and atomic commit. It is not a CPU-time measurement and is not an RTL speedup. Only a future campaign that closes model, VCS, and DC/PT on one full ledger may use that stronger label.

The mapped nine-SRAM island occupies $166,514 \mu\text{m}^2$. Synthesis and place-equivalent mapping use the foundry 28-nm standard-cell library and the nine compiler SRAM abstracts; no Yosys or educational library is used as an ASIC PPA proxy. PrimeTime reports setup/hold slack of +27.9/+1.8 ps at 3 ns under ideal clocks. Formality passes 16,549 mapped-to-mapped compare points on that netlist pair. In a separately bounded 64-row, 253-cycle directed window the island dissipates 29.08 mW and 22.07 nJ. That energy is a mixed-corner, prelayout PrimeTime PX estimate without SPEF and without a frame-level activity factor; it is not energy per DSEC frame and is not a chip measurement.

A separately sealed ep34 tile uses the first 64 real support masks of the cycle ledger. VCS matches 196 accepted issues, 58 parent edges, 31 dead-write elisions, 54/33 macro reads/writes, four forwards, six deadline holds, 14 stalls, and 64 commits/completions. Lane values are deterministic synthetic signed-12-bit data with zero prior psum. This calibrates modeled event counts and directed arithmetic; it does not itself produce the 1.6945× cycle ratio. Protocol errors, X-pessimism on energy, and failed sibling attempts in the same repository are not cited.

The 64-row task is the scheduling quantum that matches the ping-pong directory, not a claim that the deployed convolution is only 64 rows deep. Each task carries up to 16 typed sources and writes a 96-lane signed product. The nine macros are

TABLE II
FROZEN-POPULATION LOCAL DSEC ACCURACY AGAINST A HISTORICAL,
BACKEND-MISMATCHED BASELINE (825 FRAMES; LOWER IS BETTER).

Metric	Baseline	Candidate	Δ
AEE	1.199514	1.197367	-0.002147
AAE	5.400641	5.412808	+0.012167
AAE-Bench.	5.106363	5.121619	+0.015256
DSEC-Fl	5.313360	5.328834	+0.015475

TABLE III
ADMITTED C1 EVIDENCE; OPEN CLAIMS ARE NOT INFERRED.

Class	Result	Boundary
Cycle model	1.6945 $\times$; -40.99% time vs. strong zero ^[model]	Ten-sample same-ledger ratio; not an RTL speedup
Directed RTL	64-row tile: 196 issues, 58 parents, 31 dead writes, 64 commits ^[VCS]	Event-count calibration; synthetic signed-12-bit lanes
Physical	166,514 μm^2 ; +27.9/+1.8 ps; 16,549 FM pts; 29.08 mW / 22.07 nJ ^[DC/PT]	Prelayout nine-SRAM island; mixed-corner window; no SPEF

addressed as a single 1RW parent store: one grant moves either a mask, a psum, or a residual payload. Capacity eviction follows the live-parent bitmap rather than LRU of raw addresses, because a recently written parent may already be dead while an older parent remains legally observable. That replacement rule is why the concurrent-access ceiling in Table I is not a drop-in RTL mode: it would require a second port that the mapped island does not have.

The same-ledger model therefore reports two numbers that must not be conflated. 1.6945 $\times$ is C1 versus strongest-zero on the physical 1RW contract. 1.902 $\times$ is the unphysicalized ceiling. The 12.25% gap is disclosed so that a reader cannot mistake the ceiling for a second reported architecture.

For numerical binding, eight retained Conv3x3/ConvTranspose weights are quantized per output channel to narrow-range signed INT8 with a power-of-two scale. An independently checked export has zero pre-clip violations; the largest absolute-code accumulator bound for C1 is 200,219, below Acc24. An operator-local replay covers 160 full calls and 3.27168×10^9 output elements. Against FP32 it gives 0.009106 MAE, 0.012407 RMSE, 0.239204 maximum absolute error, and 0.9999789 cosine similarity. All 1,280 independently sampled Python-integer probes match; the observed final accumulator range is [-29,680, 27,619]. These are operator-local numerical results, not task AEE or a hardware speedup.

Table II uses the same 825 frames, 18 sequences, and 48,152,523 valid pixels. The candidate applies hardware-order attention to all 12 blocks and dyadic-INT8 QDQ to four C1 Conv3x3 and four decoder ConvTranspose weights. Other operators remain at checkpoint precision. The historical baseline enabled TF32/cuDNN benchmarking whereas the candidate disables both; the table is therefore a compatibility gate, not a causal accuracy improvement. Aggregate AEE falls by 0.002147, but the three auxiliary errors rise and 10 of 18 sequences regress in AEE.

IV. RELATED WORK

Prosperity isolates product sparsity from bit sparsity on PTB [1]; the latter comparison is the relevant mechanism-level analogue to Table I. Phi stores hierarchical patterns chosen by DSE [2]. We adopt their evaluation pattern—cycle simulation plus RTL/physical anchors and a baseline ladder—without relabeling their numbers as ours. FireFly-T motivates multi-spike service in a different operator [8] and is not an iso-graph baseline for C1.

SpinalFlow sorts compressed spike streams to preserve state locality, and SNE makes convolutional work proportional to the incoming event stream [4], [5]. Both establish event-driven execution. Neither removes repeated products under a finite 1RW parent lifetime with atomic completion. ODIN is a 28-nm digital neuromorphic core with online learning [3]; it is a silicon neuron/synapse fabric, not a product-capture convolution island. TCAS-II SRAM compute-in-memory briefs push sparsity into the array, including dynamic-sparsity control and PIM tutorials [6], [7]. C1 is a digital 1RW parent store with residual reconstruction, not an in-array MAC macro; we cite those briefs as the circuits-venue prior on sparsity, not as interchangeable PPA.

A 28-nm optical-flow chip jointly reports operations, external-memory traffic, latency, energy, and AEE [9]. We follow that separation of metrics and do not numerically rank its silicon measurements against this prelayout island. For a co-designed network, a published chip rarely supports the identical graph. Comparison therefore stays at three levels: same-network component baselines, iso-workload public-artifact opportunities, and published technology-normalized metrics. Results from different levels are never multiplied into a system speedup.

V. DISCUSSION

C1's circuits contribution is the 1RW contract, not a larger sparsity percentage. On this ledger, bit sparsity is already visible to a zero-skipping baseline and does not move time (1.003 $\times$). Product reuse does, but only while a legal parent occupies one of nine macros and the arbiter can still meet the row deadline. Removing either constraint returns the design to the 1.902 $\times$ ceiling, which we refuse to report as an island.

ELSA and SpikeX reuse bundled events or weights across windows [10], [11]. Those priors matter for a typed source stream, but they do not substitute for residual reconstruction of a convolution product under a live-parent bitmap. Claiming them as C1 would mix delivery reuse with product reuse. This brief keeps the two objects separate: only product reuse with atomic completion is measured here.

The physical numbers are prelayout on purpose. ZeroWireload and ideal clocks are optimistic for a nine-macro island; they are still the correct class of evidence for an Express Brief that does not have a taped-out chip. A reader looking for pJ/SOP on silicon should use ODIN or the CICC optical-flow chip [3], [9] and should not scale our 29.08 mW window to those metrics.

The ten-sample spread (1.661–1.723 $\times$) is narrow enough to argue that C1 is not a one-trace accident, and wide enough to

forbid quoting only the best sample. We therefore report the ledger aggregate $1.6945\times$ and the geometric mean $1.6946\times$ together, and we do not promote a max-sample headline. The directed 64-row VCS tile sits at the other end of that honesty rule: it is one real-mask slice used to count grants, not a second speedup.

Deadline holds and stalls in that tile (six and fourteen) are evidence that the 1RW arbiter is doing work. A design that never stalls is either dual-ported or under-occupied. C1 is neither. The 12.25% tax versus the concurrent-access ceiling is the same phenomenon at ledger scale: the port, not the parent relation, is the remaining bottleneck once dead writes are gone.

VI. LIMITATIONS

The $1.6945\times$ ratio is a cycle-model result calibrated by one real-mask RTL tile. The mapped nine-SRAM component is a separate physical anchor, not a full integration of the 240-KiB-class common-charge ledger. Energy is a directed window, not energy per frame, and the activity factor of that window is the directed 64-row mask task rather than a DSEC frame. Hold and power of a companion weight-broadcast fabric, full-network FPS, and monolithic integration remain open and are not derived from C1 ratios.

This brief therefore refuses several claims that would make a stronger headline. We do not multiply the C1 model ratio by any FC-delivery or neuron-service factor. We do not treat the concurrent-access ceiling as a second architecture. We do not report Yosys or educational-library area as ASIC PPA. We do not promote the compatibility-gate AEE delta as a quantization gain. Those omissions are completeness requirements for a binary Express Brief, not missing experiments to be inferred by the reader. If a later campaign closes matched hold and power on the weight-broadcast fabric, that result belongs in a separate manuscript with at least thirty percent new material relative to this brief; it is not an unstated extra column of Table III.

VII. CONCLUSION

The principal circuit result is not firing sparsity itself, but its executable capture under a finite single-port parent store. C1 provides bounded exact product reuse with a same-ledger model reduction of $1.6945\times$ and a 28-nm SRAM island that meets 3-ns setup/hold. Separating model, VCS, and physical evidence yields a complete Express Brief without an unsupported whole-network claim.

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